

B25 — FIT LEVEL 01 PROJECT

Intelligent Electronics Diagnostic & Assistance Workbench

A sensor-fused intelligent embedded workstation for electronics technicians and precision repair environments

Degree Programme: Bachelor of Information Technology — IT / IT & Management (ITM)

Supervisor: [Supervisor Name — to be confirmed]

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INTRODUCTION

Rethinking the Electronics Repair Bench

Electronics technicians rely on ordinary workbenches that give no feedback about safety, tool usage, or working conditions. Every repair session depends entirely on the technician noticing hazards themselves — overheating tools, poor air quality, or equipment left powered on.

This project introduces an Intelligent Electronics Diagnostic & Assistance Workbench — a sensor-fused embedded system that observes the technician's workspace in real time, automates routine actions, and actively protects both the technician and the equipment.

AT A GLANCE



Dual-controller core

ESP32 (Wi-Fi) + STM32 (real-time control)



Vision-aware

ESP32-CAM / OpenMV activity recognition



Safety-first

Air quality, thermal & current monitoring



Fully automated

Presence-driven lighting, ventilation & logging

PROJECT GOALS

Aim & Objectives



AIM

To design and develop a sensor-fused, intelligent embedded workbench that improves technician safety, automates repetitive workspace routines, and reduces human error during electronics repair and diagnostics.

OBJECTIVES

- 1 Detect technician presence and automatically activate the workspace
- 2 Identify and track tool usage using RFID and load-cell sensing
- 3 Monitor real-time power draw and thermal conditions during repairs
- 4 Detect hazardous air quality (VOC / smoke) and trigger safety response
- 5 Provide adaptive lighting based on ambient light and active work zone
- 6 Log usage data and safety events for review and accountability
- 7 Implement automatic idle / shutdown logic to conserve power

Problem & Proposed Solution



THE PROBLEM

- No safety monitoring — overheating tools and toxic solder fumes go unnoticed
- Manual tool tracking — no record of what was used or left powered on
- Static lighting — wastes power and strains the technician's eyes
- Zero automation — every routine action is manual and error-prone
- No accountability logs for incidents, tool usage, or workspace conditions

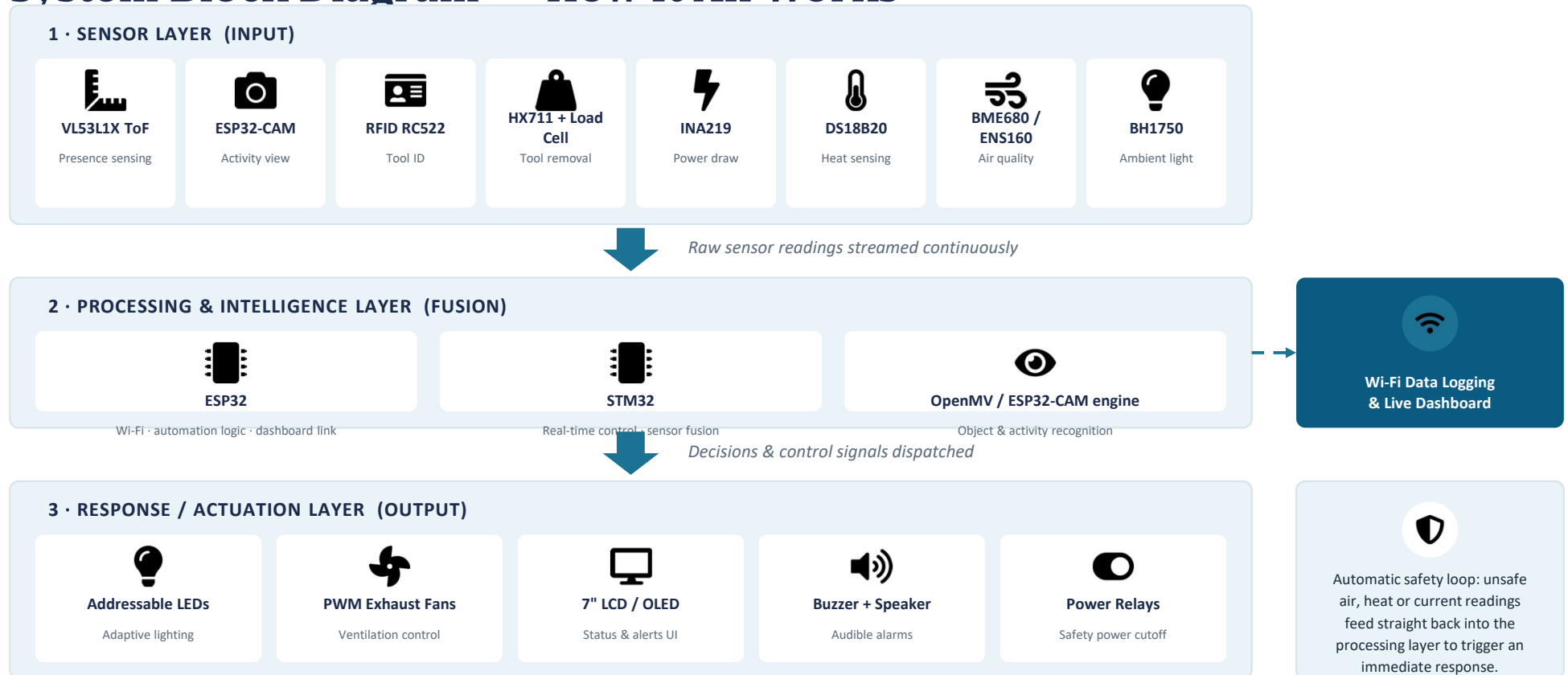


PROPOSED SOLUTION

- A sensor-fused workbench: presence, vision, RFID, load, current, thermal, air-quality and light sensors working together
- Dual-controller core (ESP32 + STM32) fuses sensor data and drives automated responses in real time
- Vision module (ESP32-CAM / OpenMV) adds activity and object awareness beyond simple sensing
- Automatic safety actions — ventilation boost, power cutoff, and alerts — without technician intervention
- All events and conditions logged for review, turning the bench into an accountable, data-driven workspace

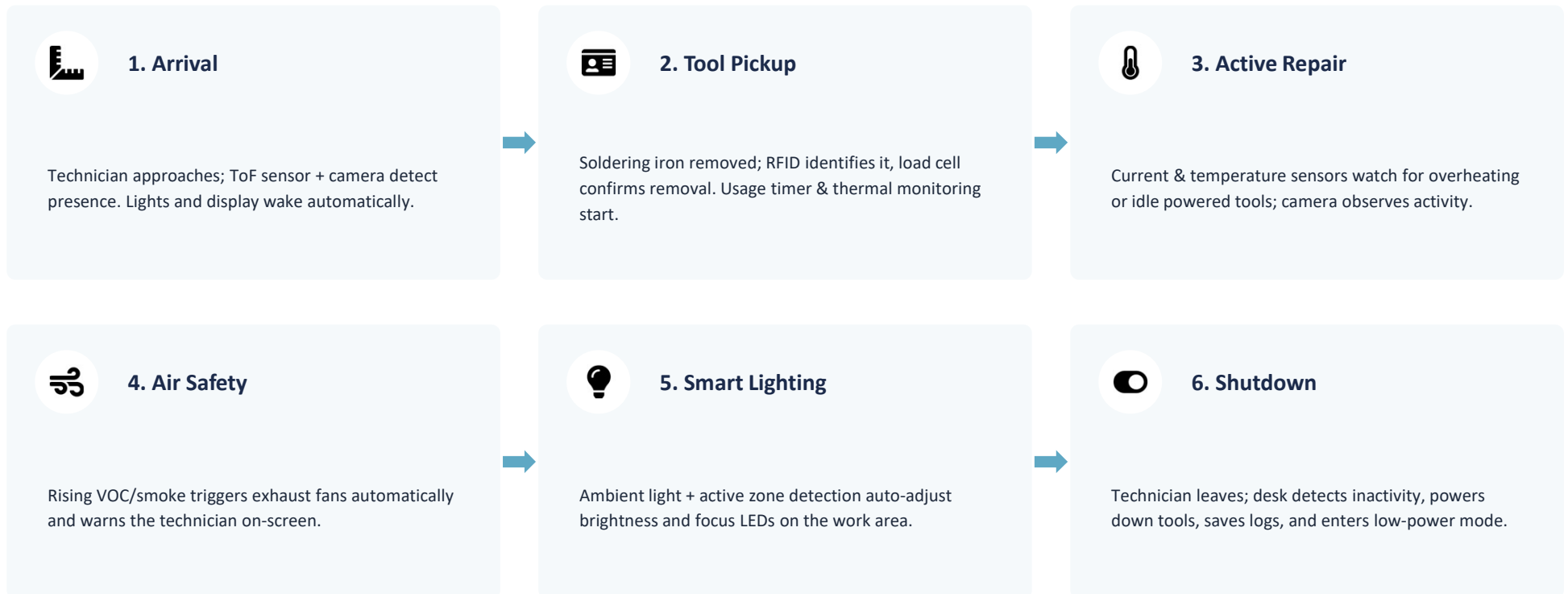
SYSTEM ARCHITECTURE

System Block Diagram — How It All Works



FROM POWER-ON TO POWER-OFF

Operational Flow — A Working Example





Hand-Drawn Concept Sketch

Space reserved for the workbench concept sketch — to be inserted here.

DIFFERENTIATION

Why This System Stands Out



Safety-first design

Real active protection — thermal, current & air-quality hazards are caught and acted on automatically, not just displayed.



True sensor fusion

8+ sensors and a vision module work together through dual controllers — not isolated add-ons bolted onto a desk.



Genuine automation

Presence, tool use, lighting and ventilation all respond on their own — no manual switches or checklists needed.



Data-driven accountability

Every session, tool use and safety event is logged, enabling review and continuous improvement.



Built for real repair work

Designed around an actual technician workflow, not a generic 'smart desk' concept.



Realistic & achievable

Every component is proven, sourceable and costed — deliverable within the project budget and timeline.

BILL OF HARDWARE

Hardware Components



ESP32 Dev Board

Wi-Fi, automation logic, dashboard link



STM32 Board

Real-time control & sensor fusion



ESP32-CAM / OpenMV Cam

Vision & activity recognition



VL53L1X ToF Sensor

Presence / distance sensing



BME680 / ENS160

Air quality (VOC) sensing



DS18B20

Thermal monitoring



INA219 / INA226

Current & power monitoring



BH1750

Digital ambient light sensing



RFID RC522 + Tags

Tool identification



HX711 + Load Cell

Tool weight / removal detection



7" LCD / OLED Display

System feedback UI



Addressable LED Strips

Adaptive workspace lighting



PWM Exhaust Fans

Smart airflow / ventilation



Buzzer + Speaker

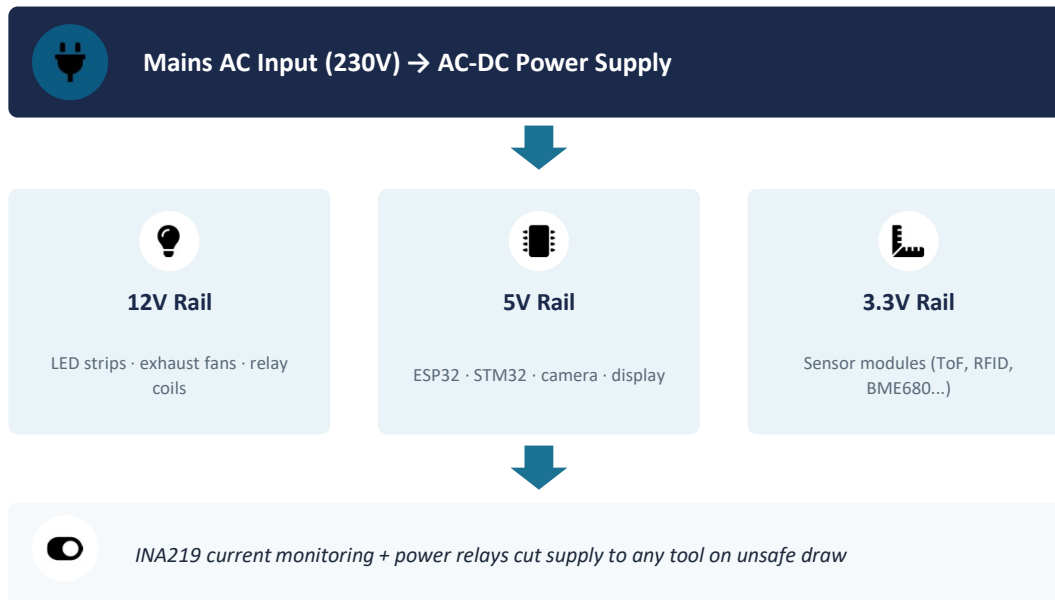
Audible alerts & warnings



Power Supplies & Relays

Distribution & safety cutoff

Power System



DESIGN NOTES

- Logic and sensor rails are electrically isolated from motor/LED rails to avoid noise on sensitive readings.
- Relays provide hardware-level safety cutoff, independent of software, for overcurrent or overheating events.
- PWM control on fans allows variable-speed ventilation instead of simple on/off switching.
- A fused, regulated supply protects the ESP32/STM32 core from voltage spikes on the 12V rail.

THE SOFTWARE STACK

Software Components & Technologies



Firmware

- C / C++ (Arduino framework) on ESP32 & STM32
- Sensor libraries: Adafruit BME680, VL53L1X, HX711, MFRC522
- FreeRTOS-based task scheduling on STM32



Vision Processing

- OpenMV IDE / MicroPython for the vision module
- Frame-differencing & object detection for activity awareness



Connectivity & Dashboard

- Wi-Fi (ESP32) with MQTT / HTTP for telemetry
- Node-RED or a lightweight web dashboard for live monitoring
- Firebase / SQLite for event & usage logging



Development & Design Tools

- Git / GitHub for version control
- KiCad for PCB & schematic design
- Fritzing / Proteus for circuit prototyping

COST PLANNING

Detailed Project Budget

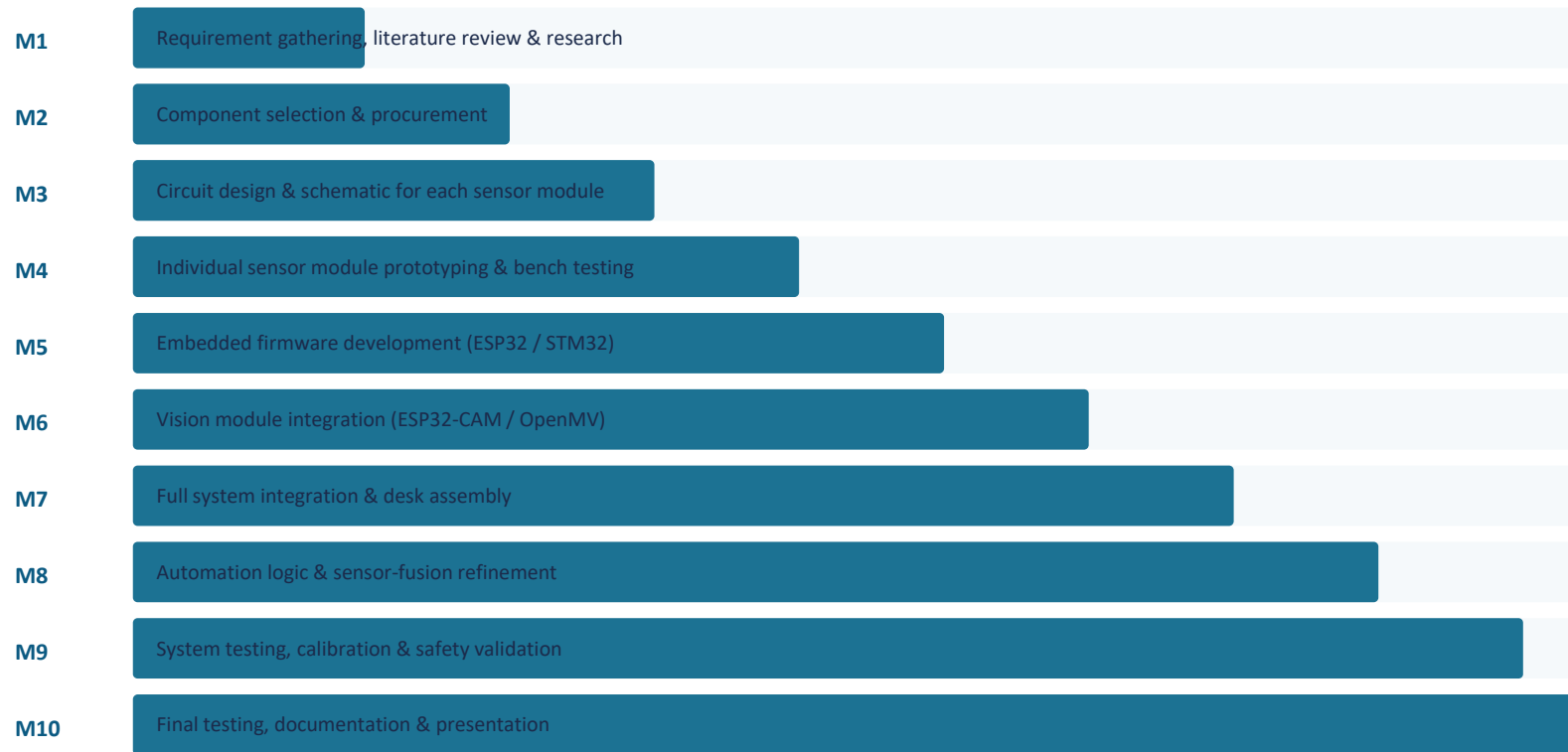
Component	Estimated Cost (LKR)
ESP32 development boards	5,000
STM32 board	6,000
ESP32-CAM / OpenMV camera	10,000
VL53L1X ToF sensors	5,000
BME680 / ENS160 air quality sensors	5,000
INA219 current sensing modules	3,000
DS18B20 temperature sensors	2,000
BH1750 light sensors	2,000
RFID system (RC522 + tags)	4,000
Load cells + HX711 modules	5,000
7-inch LCD / OLED display	10,000
LED lighting system	5,000
Cooling / exhaust fans	4,000
Power supplies & relays	5,000
Wiring / PCB / connectors	7,000
Desk / frame fabrication	16,000

TOTAL ESTIMATED BUDGET

LKR 94,000

10-MONTH PLAN

Project Timeline



TEAM RESPONSIBILITIES

Work Distribution

Every member owns at least one sensor module end-to-end — from circuit design to firmware — with a focus on IoT & hardware integration.

 IT	 IT	 ITM	 IT	 ITM
CHAMIDYA J.A.S 254037B	DE SILVA P.S.S 254048J	OMESH N.A.N 255078H	KALANSOORIYA O.K 254101R	GOONATILLAKA W.A.D.S.S 255027D
Presence & Vision Module	Tool Tracking Module	Air Quality & Ventilation	Power & Thermal Monitoring	Lighting & Display Module
VL53L1X ToF sensor + ESP32-CAM: presence detection, workspace-region recognition & wake logic	RFID (RC522) + HX711 load cell: tool identification, pickup/removal detection & usage timers	BME680/ENS160 air sensor + PWM exhaust fan control for automated ventilation safety	INA219 current sensor + DS18B20 thermal sensor, plus the power distribution & relay safety circuit	BH1750 light sensor + addressable LEDs + LCD/OLED display UI, and documentation coordination

SOURCES CONSULTED

References

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Thank You

Questions & Discussion Welcome